

MEHDI BELHAJ

Power Electronics & Hardware Engineer

Analog & digital electronics · Embedded systems · Power conversion · Modelling & control · Aerospace qualification — Toulouse & Europe

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270 VDC

SiC 1200 / 1700 V

10 kW

DO-160G · DAL A

EMC / EMI

OrCAD · KiCad

Matlab · Simulink · Simscape

PI / PID controllers

PSIM · LTspice

C · C++ · Python

AVAILABILITY

Immediate

EXPERIENCE

3 years · aerospace

MOBILITY

Toulouse · France · Europe

WORK PERMIT

Valid to Nov. 2028

DRIVING LICENCE

B

ENGLISH

C1 · fluent

CAREER OBJECTIVE

Join an engineering team on a **long-term permanent contract**, ideally as **technical lead** in power electronics, analog electronics or control, to design high-criticality embedded boards and systems covering the full V-cycle, from specification to qualification testing. Primary target: aerospace and space in the Toulouse area; open to defence, energy, rail, electric mobility and industry, in France and abroad. Longer term, move towards **embedded systems** and software (C / C++ / Python), including **embedded AI**.

PROFILE SUMMARY

Power electronics and hardware engineer with **3 years of experience in the aerospace sector** at Airbus, Safran Ventilation Systems and SuperGrid Institute. Specialised in the complete design of high-voltage embedded boards: **Flyback 540 V**, **Buck**, **SEPIC** and **LDO** supplies, three-phase **SiC 1200 V inverter up to 10 kW**, **DO-160G** EMC filtering (Section 21, categories B and H), analog and digital sensing chains, up to **DAL A** criticality.

Dual hardware and control competence: electrical and electromechanical equations, block-diagram modelling in **Matlab / Simulink / Simscape**, control-law design and **PI / PID controller tuning**, closed-loop simulation and stability analyses (Bode, Nyquist, Nichols, margins) with Control System Toolbox and Simulink Control Design.

Key differentiator: autonomous coverage of the V-cycle as **author of the key deliverables** — design justification file, PCB layout specification, technical requirements specification, hardware verification procedure, test plans, compliance matrix — and active across the industrial process: Windchill PLM, codification and bills of materials, supplier consultations, obsolescence, multidisciplinary meetings and gate reviews.

Profile broadened by **330 h of Cisco / OpenEDG certificate training** in C++ and Python (2025-2026) and by an **entrepreneurial activity** run in parallel (CE, RoHS, WEEE, EPR and UKCA product compliance, sourcing, business management). Bilingual French, native Arabic, English C1 (weekly technical meetings with Safran USA).

KEY STRENGTHS

- **Autonomy across the V-cycle** — sole electronics lead on my last two assignments, from kick-off to testing, author of all technical deliverables.
- **Dual power & analog culture** — supply chains (Flyback, Buck, SEPIC, LDO), SiC 1200 / 1700 V stages up to 10 kW, sensor conditioning and 24-bit acquisition.
- **Modelling & control** — equations, Simulink / Simscape block diagrams, PI / PID design and tuning, closed-loop simulation, small-signal stability; 130 ms dynamic requirement validated at Airbus.
- **EMC and aerospace qualification** — common- and differential-mode filtering, lightning, ESD, DO-160G campaigns run end to end, dielectric tests up to 20 kV.
- **Project management and supply chain** — multidisciplinary meetings, gate reviews, codification and BOM, supplier consultations, obsolescence and second sourcing, Windchill PLM.
- **Software & data openness** — certified C++ and Python (330 h), oriented towards real-time embedded software, test automation and measurement post-processing.
- **Immediate availability & mobility** — available without notice, mobile in Toulouse, France and Europe.

DETAILED TECHNICAL SKILLS

POWER ELECTRONICS

Three-phase **SiC 1200 V / 1700 V — 10 kW** pulsed inverter · Quasi-resonant Flyback 540 V to 24 V without optocoupler · Isolated multi-output Flyback 28 V to 3.3 V · Buck · SEPIC · LDO · Isolated gate drivers UCC21750 (DESAT, Miller clamp) · Driver supply and biasing · RCD snubbers · DC-bus pre-charge and inrush limiting · Magnetic component sizing

ANALOG ELECTRONICS & SENSING

Current sensing (ACS37002, INA190, shunts) · High-voltage sensing (resistive dividers and MAX1363 ADC) · Temperature (TMP1075) · 24-bit force measurement chain (ADS1247) · Position interface (CAM312) · Operational amplifiers (OPA387) · Signal conditioning and filtering · Galvanic isolation (ISO1640)

DIGITAL & EMBEDDED

TMS320F280049 DSP (C2000) · PIC16F877x · Embedded C · **C++** (OOP, templates, STL) · **Python** — 330 h certified · VHDL (basics) · LabVIEW · UART, SPI, I2C, RS422, PWM, ADC, EEPROM · Contribution to **DAL A** embedded software (equipment software, sequencer, BITE, sub-assembly specifications) · Git · Linux · PyTest

MODELLING, CONTROL & SIMULATION

Matlab / Simulink / Simscape · Control System Toolbox · Simulink Control Design (automated linearisation: sLinearizer, getIOTransfer, getLoopTransfer, getSensitivity) · Scilab / Xcos · Electrical and electromechanical equations (converters, DC bus, motor, actuator) · Transfer functions and **block-diagram modelling** · **Control-law** design · **PI / PID controller** design and tuning (identification, pole placement, margin criteria) · Nested current / voltage loops · **Closed-loop simulation**: load steps, field weakening · Small-signal stability (open-loop transfer functions, sensitivities, gain, phase, delay and modulus margins) · Bode, Nyquist, Nichols · PSIM (FFT) · LTspice · Saber

EMC, PROTECTIONS & TESTING

DO-160G Section 21 — categories B and H: conducted and radiated emissions, LISN measurements · Common- and differential-mode filters · Lightning protection by pin injection · 15 kV ESD · Reverse-polarity protection · Phase-loss detection on ± 270 VDC · Over-voltage and over-temperature · Dielectric tests up to 20 kV · Insulation measurements · IEC 60068 · ABD0100 requirements · Bench debugging, oscilloscope

CAD & PCB DESIGN

OrCAD Capture CIS · KiCad · Proteus · **IPC-2221** layout specifications (high-voltage clearances corrected for altitude per IEC 60664-1) · 6-layer FR4 PCB, stack-up · Current-based trace sizing · DFX and testability · EMC and thermal constraints · Mechanical integration on 3D model · Flex circuits · IPC-A-600, IPC-A-610, IPC-7351

PROCESS, QUALITY & INDUSTRIALISATION

Full V-cycle · Internal and gate reviews · Authoring: technical requirements specification, hardware verification procedure, design justification file, calculation notes, test plans, compliance matrix, lessons-learned sheet · **Windchill PLM**, codification and bills of materials · Supplier consultations · Aerospace component qualification · Obsolescence and second sourcing · **DO-254** (basics) · CE, RoHS, WEEE, EPR, UKCA product compliance · International customer interface in English

DEVELOPMENT METHODOLOGY

Specification

Requirements · TRS · DO-160 · ABD0100

Architecture

Block diagram · topologies · budgets

Design

Schematics · DJF · modelling · layout · BOM

Build

Prototype boards · procurement

Unit tests

Voltages · insulation · bench debug

Integration

EMC · lightning · ESD · thermal

Validation

Compliance matrix · capitalisation

DETAILED PROFESSIONAL EXPERIENCE



Power Electronics & Hardware Engineer — Safran Ventilation Systems

Via Centum T&S · Toulouse (Blagnac) · Electronics lead · Embedded 270 VDC power conversion · DO-160 category H

Nov. 2023 — May 2025 · 19 months

CONTEXT

Adaptation of an existing aircraft ventilation system from the 115 VAC network to the **270 VDC** high-voltage network (more-electric aircraft and helicopter), to meet the needs of future helicopter programmes. Goal: deliver a commercially oriented demonstrator while minimising software and mechanical impacts. Electronics lead role with the V-cycle covered autonomously.

ACHIEVEMENTS

- **Architecture & hardware design**: complete design of the **input board** (EMI filter and power supplies) within a multi-board architecture (input, inverter control, interface, power); 1,890 W delivered, -40 to +100 °C temperature range.
- **Nine hardware functions** designed and justified: diode reverse-polarity protection; low-voltage supplies (isolated Flyback, 18 / 16 / 5 V LDOs, Buck converter) with separate rails for the high- and low-voltage groups; **phase-loss detection in under 4 ms** with supply hold-up over micro-cuts of 200 ms and more; inrush limiting to a normative envelope (9x / 4x / 2x nominal current); lightning protection by pin injection; 15 kV ESD; category H EMI filter; over-temperature protection; temperature measurement.
- **PCB design & layout**: layout specification — **6-layer FR4 PCB** (35 μ m, 1.6 mm, class 4, selective NiAu finish), stack-up definition, **IPC-2221** clearance calculations corrected for altitude per IEC 60664-1, DC-bus traces rated for 9 A RMS ($\Delta T < 10$ °C), Y-shaped lightning routing on outer layers, DFX rules and testability (100 % of nets testable).
- **Mechanical integration**: interface drawing, heat-exchange zones, connector placement, iterations with the mechanical design office on the 3D model.
- **EMC qualification**: DO-160G category H — common- and differential-mode filter sizing, RCD snubbers, PSIM simulations and FFT analyses; EMI test plans written and executed (LISN, conducted and radiated emissions); lightning tests and insulation measurements.
- **Industrial documentation**: **design justification file** with per-function justification and calculation notes, **technical requirements specification**, **hardware verification procedure**, **requirements verification matrix** and lessons-learned sheet.
- **Configuration management, PLM & procurement**: component codification and bills of materials in Windchill; supplier consultations and selection, aerospace component qualification (temperature, altitude, derating), obsolescence and second sourcing.
- **Coordination**: multidisciplinary technical meetings (hardware, layout, mechanical, procurement), internal and gate reviews, weekly technical meeting led in English with the Safran USA teams.

Results: functional mock-up validated in EMI tests on the applicable DO-160 sections; complete design package delivered; 270 VDC feasibility demonstrated with minimised software and mechanical impacts, work capitalised for future 270 VDC programmes.

Environment: OrCAD Capture CIS · KiCad · PSIM · LTspice · Matlab/Simulink · TMS320F280049 · DO-160G cat. H · IPC-2221 / A-600 / A-610 / 7351 · IEC 60664-1 · IEC 60068 · Windchill PLM

CONTEXT

Proof of concept of an ultra-fast electromechanical actuator ("Pop-Up Spoiler" EMA) deploying a wing surface in **130 ms** or less, to counter gusts and lighten the wing structure, within the Zero Emission aircraft studies. Design of the motor-control electronics: three boards — 28 V CPU, ± 270 V HVDC and inverter — in a constrained $65 \times 65 \times 170$ mm volume. Partnership between Serma Ingénierie, Crouzet and Airbus.

ACHIEVEMENTS

- **Requirements, specifications & safety:** contribution to the hardware and software requirements and to the safety assessment carried out with Serma Safety; application of ABD0100 aerospace requirements.
- **SiC driver board:** supply, biasing and gate driving of 6 SiC 1200 V MOSFETs — isolated UCC21750 drivers with DESAT protection, four driver supplies, phase-current and temperature sensors; three-phase inverter delivering **10 kW in pulsed operation**.
- **HVDC board:** ESD protection and inrush limiting, 540 V EMC filter (common and differential modes, 900 V ceramic capacitors), high-voltage current and voltage sensing (ACS37002 and MAX1363 ADC), **quasi-resonant Flyback 540 V to 24 V without optocoupler** (auxiliary-winding regulation) and Buck 24 V to 12 V for the brake and locking device (TPS51604, INA190 sensing).
- **CPU board:** 28 V EMC filter, ESD and reverse-polarity protection, dual-output Flyback 28 V to 3.3 V analog / digital (LM5155), 5 V converter, force acquisition chain with a **24-bit ADS1247 ADC**, CAM312 position interface, **TMS320F280049 DSP** and peripherals (UART, SPI, I2C, RS422, PWM, ADC).
- **Modelling & control laws:** electrical and electromechanical equations of the sub-assemblies (motor, power electronics, mechanical chain), **block-diagram models in Matlab / Simulink / Simscape** and Scilab / Xcos, parameter identification; control-law design and **controller tuning**, closed-loop simulation and validation of the 130 ms dynamic requirement (Bode and Nyquist analyses); power-electronics simulations in LTspice and PSIM (PWM, losses, transients).
- **DAL A embedded software:** contribution to the preliminary and detailed design of the equipment software, sequencer and built-in tests (BITE); specifications of the electronic sub-assemblies.
- **Reliability:** junction-temperature analysis of the SiC components in pulsed operation (thermal shocks linked to the 130 ms sequences), ageing risks, reliability reviews with the partners.
- **Integration & reviews:** board-to-board flex links, dedicated driver-supply board, work on the 3D model with the mechanical engineers; preliminary and detailed design justification files; design reviews with Airbus and Crouzet.

Results: 130 ms requirement validated in simulation; hardware architecture approved in Airbus and Crouzet reviews; proof of concept integrated into the Zero Emission studies; know-how capitalised on transient control.

Environment: OrCAD · KiCad · LTspice · PSIM · Matlab / Simulink / Simscape · Scilab / Xcos · SiC 1200 V · UCC21750 · BM25C123FP2 · LM5155 · TPS51604 · MAX1363 · INA190 · ADS1247 · CAM312 · ACS37002 · TMP1075 · DAL A · ABD0100 · DO-160G Section 21



CONTEXT

Research and development on high-voltage DC grids: design of an isolated high-voltage supply for the gate drivers of 6 series-connected 1.7 kV SiC MOSFETs in a high-voltage leg.

ACHIEVEMENTS

- **Design:** isolated HV supply based on 12 cascaded ferrite transformers driving 6 series-connected 1.7 kV SiC MOSFETs (high and low side); TDK EPCOS T38 core selection balancing losses, insulation and footprint.
- **Simulation & correlation:** LTspice modelling of the transformers' magnetic and electrical behaviour, validated by oscilloscope measurements.
- **EMC analysis:** characterisation of common-mode currents generated by edges up to $16 \text{ kV}/\mu\text{s}$ and definition of countermeasures.
- **Laboratory testing:** prototyping and dielectric tests up to 20 kV to validate insulation and safety.
- **Industrialisation:** specification for industrial prototyping (efficiency, EMC compatibility).

Results: 20 kV insulation validated, 12-transformer cascade operational, industrialisation specifications delivered to the prototyping team.

Environment: LTspice · Oscilloscope · 20 kV dielectric tests · TDK EPCOS T38 ferrites · 1700 V SiC MOSFETs · EMC

CONTINUING EDUCATION — SOFTWARE DEVELOPMENT



CONTEXT

Structured software upskilling (330 h) completed in 2025-2026 alongside the entrepreneurial activity, oriented towards real-time embedded software and test tooling.

ACHIEVEMENTS

- **C++ Advanced (CPP):** function and class templates, full STL (containers, algorithms, sorting, merging, sets, iterators), stream, file and string I/O.
- **C++ Essentials 1 & 2 (CPE, CPA):** types, operators, flow control, functions; pointers and dynamic allocation; complete object-oriented programming (inheritance, polymorphism, abstraction); exceptions, namespaces, streams.
- **Python Essentials 1 & 2 (PCEP, PCAP):** algorithms, script design and debugging; object-oriented programming, exceptions, iterators and generators, modules, packages and PIP, file handling.

- **Target applications:** real-time embedded software in C / C++, test automation and measurement post-processing in Python (test benches, EMI campaigns, oscilloscope acquisitions).
- **Published projects** — github.com/mehdi-belhaj-eea: scope-emi-toolkit (oscilloscope CSV parser, dBμV spectrum, EMC pre-check, 8 unit tests, continuous integration) · embedded-dsp-cpp (ring buffer and digital filters in C++17, 12 GoogleTest tests).

Results: 5 verifiable Credly badges, CPP / CPA / CPE and PCAP / PCEP frameworks.



ENTREPRENEURSHIP — IN PARALLEL

TOKYOSEAT

Co-founding partner — TOKYOSEAT
Electrical and sanitary equipment · SAS with 5 partners · trademark registered at INPI · Amazon FBA and Shopify (France, Germany, Italy, Spain, Netherlands) and United Kingdom

Since Jul. 2023 · ongoing

CONTEXT
Creation and development of a European brand of electrical and sanitary home equipment, distributed mainly on European marketplaces. Founding-partner role covering product compliance, sourcing, supply chain and business management. The business is now stable, with day-to-day operations run by an employee: compatible with a full-time engineering commitment.

- ACHIEVEMENTS**
- **Product & regulatory compliance:** CE marking under the Low Voltage and EMC directives, RoHS, WEEE, extended producer responsibility (EPR) and UKCA — laboratory test coordination (electrical safety, EMC), preparation and verification of technical files and EU declarations of conformity with French and German bodies, multi-country regulatory watch.
 - **Environmental compliance & labels:** WEEE and packaging EPR registrations in France and Germany (Stiftung EAR register); German quality label “SEHR GUT (1.3)” obtained on the flagship product (DasQualitätssiegel.de, independent product tests, October 2025).
 - **Supplier & product quality:** factory selection, negotiation, sampling campaigns, quality control, improvement plans built from customer feedback; filing and monitoring of the TOKYOSEAT trademark at INPI (classes 9 and 11).
 - **Operations & automation:** Amazon FBA (pan-European programme) and Shopify store backed by a third-party fulfilment centre (storage, order preparation, delivery); import (EORI, TARIC classification, customs, VAT, forwarders); automation of order flows and customer service, API integrations, automated report and KPI analysis.
 - **Business management:** business plans, cost price, margins, ROI, pricing scenarios, cash flow, profitability dashboards per SKU; recruitment and management of one employee.

Results: compliance maintained on 6 markets (EU and UK) without interruption of sales; German quality label obtained; German and British markets developed in 2025-2026; business stable and run day to day by an employee.

KEY MEASURABLE ACHIEVEMENTS

PROJECT	MISSION	RESULT / IMPACT
270 VDC conversion Safran SVS · 2023-25	Input board, 9 protection functions, layout specification, DO-160G cat. H qualification, full documentation, procurement and codification.	EMI-validated mock-up · complete package delivered · V-cycle covered autonomously
EMA actuator Airbus · 2022-23	3 boards within 65 × 65 × 170 mm: 10 kW SiC inverter, 540 V Flyback, Simulink / Simscape modelling and control laws, DAL A software, safety.	130 ms requirement validated · architecture approved in Airbus and Crouzet reviews
Isolated HV SiC supply SuperGrid · 2022	12 cascaded ferrite transformers for 6 series-connected 1.7 kV SiC MOSFETs, EMC analysis at 16 kV/μs.	20 kV insulation validated · industrialisation specifications delivered
TOKYOSEAT since 2023	CE, RoHS, WEEE, EPR and UKCA compliance; sourcing and supplier quality; import and supply chain; business management.	6 compliant markets · “SEHR GUT (1.3)” label
Software training Cisco / OpenEDG · 2025-26	330 h: C++ Advanced, C++ Essentials 1 and 2, Python Essentials 1 and 2.	5 verifiable Credly badges

EDUCATION & CERTIFICATIONS

 	Master's degree in EEA (5-year degree) • Université Toulouse III Paul Sabatier • INP Toulouse – ENSEEIHT Electronics, Electrical Energy, Automatic Control • Electrical Energy track: Conversion, Materials, Sustainable Development • specialisation in Power Electronics, Actuators and Control	2020 – 2022
	Bachelor's degree in EEA — Electronics, Electrical Energy, Automatic Control • Université Toulouse III Paul Sabatier	2019
	Two-year technical degree (DUT) in Electrical Engineering and Industrial Computing • ENSAM Rabat, Morocco	2018
	Cisco Networking Academy / OpenEDG — C++ and Python C++ Advanced, C++ Essentials 1 and 2, Python Essentials 1 and 2 • verifiable Credly badges	2025 – 2026 • 330 h
	French electrical safety authorisation H0, B2V, B2V Essai, BE Essai — CAMIRA Formation Prévention Non-electrical work near HV installations (H0), work and testing on LV installations (B2V, BE Essai) • refresher due	2022
  	EYE Berlin — European Youth Event • European Parliament & Schwarzkopf Foundation • certificate of participation Berlin, 18-20 April 2024: 1,500 young Europeans gathered for workshops and debates on the European elections and the future of a sustainable, innovative and united Europe	Apr. 2024

LANGUAGES & SOFT SKILLS

French — bilingual • **Arabic** — native • **English** — C1, fluent: weekly technical meetings with Safran USA, documentation and deliverables in English • **Spanish, Portuguese** — basics

Autonomous • rigorous • curious • responsive • team player • proactive • technical leadership • Competitive football • NQT mentoring • travel — 18 countries visited

CONDITIONS & AVAILABILITY

Availability: immediate, no notice period • **Contract:** permanent, fixed-term, project-based or temporary — full time

Mobility: Toulouse, Occitanie, all of France, Europe — occasional travel accepted • **Driving licence:** B

Status: Moroccan national, French salaried residence permit valid until November 2028

Target sectors: aerospace and space, defence, energy, rail, electric mobility, semiconductors, industry

Skills portfolio — Mehdi Belhaj • document prepared for recruitment purposes: all information corresponds to assignments actually carried out • References available on request • Updated — October 2026